

The Hand That Gives

*Pre-PIE *de- and the Polyfunctional Architecture: the Body-Part Noun *de-r-, the Heteroclitlic Gift-Noun *deh₃-wr_̥-, and the Morphological Skeleton of the der- Project*

Paper 1 of the der- project

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Abstract

A comparative cluster first assembled by Pokorny — Greek δῶρον ‘gift; palm-breadth,’ Hesychian δάρις ‘span of all fingers,’ Celtic *durno- ‘fist, hand,’ and Latvian dūre ‘fist’ — has resisted formal etymological analysis for over a century. Schrijver (1995) describes the δωρ-/δαρ- relation as ‘obscure’; Matasóvič (2009) finds no satisfactory Indo-European etymology for Celtic *durno- and classifies it as probably non-IE in origin. Neither assessment is unreasonable given the evidence available, but a unified account is available once the cluster is recognized as reflecting two parallel derivational outputs on a single pre-PIE element *de- ‘hand’: a locative-extended body-part noun *de-r- (yielding δάρις, Old Irish derna, Continental Celtic *DERNV[]) and a heteroclitlic gift-noun *deh₃-wr_̥-/-w(e/o)n- (yielding δῶρον, Latin dōnum, Celtic *durno-, Latvian dūre). The Celtic u that has frustrated derivation is the regular reflex of a Proto-Nuclear-IE metathesis *wr_̥ > *ru > *ur, independently reconstructed by Clayton (2023) for the *-wr_̥-/-w(e/o)n- heteroclitic class. This paper also sets out the morphological skeleton for the broader four-paper project. Probability: 30–40%.

1. Introduction

The comparative cluster under discussion here has been recognized as a unit since Pokorny assembled it at IEW 203: Greek δῶρον ‘gift; palm-breadth,’ Hesychian δάρις ‘span of all fingers,’ Old Irish dorn, Welsh dwrn, Breton dorn/dourn, Continental Celtic Durnomagus, Durnacos, Dagodurnus, and Latvian dūre / dūris ‘fist.’ The semantic coherence of the set — hands, fists, palms, spans, and the gifts that hands convey — suggests a common origin. The formal relationships, however, have remained resistant to systematic analysis.

Three specific difficulties have stood in the way. The first is the vowel alternation within Greek: δῶρον has ω where δάρις has α, and no standard reconstruction has explained this within the laryngeal framework to general satisfaction. Schrijver (1995: 58), listing the forms as ‘possible

cognates,’ described the δωρ-/δαρ- relation as simply ‘obscure’ — a characterization that accurately reflects the state of the evidence rather than any oversight on his part.

The second difficulty is the consistent u vocalism in Celtic. The relevant forms appear not only in Brythonic — where a Welsh-internal raising rule o > u/___r might provide a partial account — but in Continental Gaulish from the first century BCE: Durnomagus, Durnacos, Dagodurnus. Ball and Müller (2009) confirm that the raising rule is Welsh-internal only; the u is original and must be explained at the Proto-Celtic level or above. Beekes, noting that Celtic ‘points to PIE *u, which excludes any relationship with δῶρον,’ concluded that the Celtic and Baltic forms are simply not cognate with δῶρον and separated the two senses of δῶρον itself into unrelated entries. This is a reasonable response to a genuine formal problem.

The third difficulty is the semantic gap: Celtic and Baltic preserve the ‘hand/fist’ meaning while Greek and Latin preserve the ‘gift’ meaning. The semantic connection between giving and the hand is natural enough — gifts are conveyed by hand, and Vitruvius (De architectura 2.3.3) explicitly connects the two senses of δῶρον on exactly these grounds — but a natural semantic connection is not the same as a formal etymological account. It was Matasóvič (2009, s.v. *durno-) who drew the appropriate conclusion: without a viable IE etymology, Celtic *durno- should be classified as probably non-IE in origin. The substrate hypothesis is not a counsel of despair; it is a disciplined response to the absence of a satisfactory formal account.

This paper proposes that a unified formal account is available. The cluster reflects two parallel derivational outputs on a single pre-PIE element *de- ‘open hand.’ A locative extension *de-r- produces the body-part noun attested in Hesychian δάρυς, Old Irish derna, and Continental Celtic *DERNV[. A heteroclitic extension on the verbal stem *deh₃-wr₃-/-w(e/o)n- produces the gift-noun family attested in δῶρον, Latin dōnum, Celtic *durno-, and Latvian dūre. The δωρ-/δαρ- alternation resolves as the strong/weak stem contrast of the heteroclitic paradigm. The Celtic u is the regular outcome of a Proto-Nuclear-IE metathesis, independently reconstructed by Clayton (2023) for the *-wr₃-/-w(e/o)n- class. The semantic gap dissolves because both branches descend from the same hand-noun, one preserving the body-part meaning and one preserving the function of giving.

The paper is organized as follows. §2 sets out the polyfunctional architecture of the broader four-paper project, of which the present lexical etymology is the first component. §3 develops the body-part noun *de-r-. §4 develops the heteroclitic gift-noun family, drawing on Clayton’s (2023) reconstruction of the *-wr₃-/-w(e/o)n- heteroclitic class. §5 traces the Celtic *durno- through the metathesis pathway. §6 returns to the δωρ-/δαρ- alternation and the Vitruvius testimony. §7 engages previous accounts. §8 notes the Anatolian evidence and its bearing on the chronological

architecture. §9 sets out the proposal’s scope and limitations. §10 assesses probability.

2. The Polyfunctional Architecture

The proposal rests on the reconstruction of a pre-PIE element *de- ‘hand’ that was productive across multiple derivational pathways simultaneously. Body-part terms are cross-linguistically among the most fertile sources of grammaticalization and morphological productivity: the hand reaches, points, conveys, spans, and gives, and these functions leave distinguishable lexical and grammatical residues. Heine and Kuteva (2002) document the resulting range of pathways — HAND > LOCATIVE, HAND > AGENT MARKER, HAND > GIVE — across typologically diverse languages. The present proposal is that pre-PIE instantiated several of these pathways in parallel, and that the comparative evidence in IE reflects the different outputs.

The full architecture of the four-paper project is shown in Table 1. The present paper develops the first two rows; Papers 2–4 develop the remaining branches. The reconstruction does not require any new PIE roots: the bare *de- and the locative *-r- are independently attested morphemes, and the framework is a proposal about their prior productive relationship, not an introduction of new material.

Root + suffix	Surface form	Meaning	Paper
*de- + *-r-	*de-r-	body-part noun (δάρις, derma)	1
*de- + *-u-r̥-/-(e/o)n-	*deh ₃ -w _{r̥} -	gift-noun (δῶρον, dōnum, *durno-)	1
*de- + *-r- + *-o-	*(t)ero-	bilateral-contrastive suffix	2
*de- + *-u- + *-o/-i-	*du-o-/du-i-	numeral ‘two’	2
*de- + *-h ₃ -	*deh ₃ -	give-verb (post-Anatolian)	3
*de- + postposed *-a	*de + a → *dā-	‘take’ (Anatolian)	4

Table 1. Derivational pathways on pre-PIE *de- across the four-paper project.

Two features of the skeleton bear on the present paper’s central claims. First, the locative *-r- that appears in both the body-part noun (row 1) and the contrastive suffix (row 3) has independent support from three published treatments: Lipp (2009), reconstructing the PIE heteroclitlic r/n-stem paradigm, treats *-r and *-én as parallel locative endings; Bauhaus (2019) argues that *-r is a true PIE locative case ending rather than an adverbial particle; and Kahl (2024), working on degree morphology, independently derives the secondary comparative of Indo-Iranian and Greek

from ‘an archaic locative morpheme *-er.’ Second, the bare *de- directional particle is not a theoretical abstraction: it survives as the Greek allative clitic -δε (productive in Mycenaean by 1400–1200 BCE; Lejeune 1961, van Hes 2021), as Old Church Slavonic do, as Proto-Germanic *tō (English to, German zu), and in Celtiberian (Beltrán Lloris et al. 2021; Prósper 2022: 19 n.45). The framework treats these as the grammaticalized descendants of a pre-PIE hand-element, following the well-documented pathway HAND > DIRECTIONAL (Heine & Kuteva 2002).

3. The Body-Part Noun *de-r-

The locative-extended body-part noun surfaces most directly in Hesychian δάρις / δάριρ, glossed as ‘span of all fingers’ (Attic σπιθαμή). The form is an i-stem derivative of *de-r-, morphologically transparent: the open palm as a locative body-part anchor, ‘the span at the place of the hand-direction.’

Old Irish derna (Gaelic deàrna) ‘palm of the hand’ continues the unmarked feminine singulative *de-r-nā-. The Bronze of Novallas — a Celtiberian hospitium bronze edited by Stifter (2022) — preserves Continental Celtic *DERNV[alongside the directional particle DV (Paper 3’s Celtiberian reflex of *de/do). Two derivational branches of the bare hand-element appear in a single inscription, a convergence that the existing literature does not note, since it is visible only through the framework’s lens.

The locative status of *-r- is central to the derivation. Lipp’s argument concerns the locative singular forms of acrostatic heteroclite neuters; Bauhaus’s is broader and covers pronominal and nominal formations across the family. His phrasing is direct: *-r is ‘a real case ending, not an Adverbialendung.’ Kahl reaches the same conclusion through degree morphology. The convergence of three independent treatments from three scholarly traditions is the strongest available warrant for treating *-r- as a productive locative morpheme in early PIE. The compositional reading of *de-r- — ‘at the place of the hand-direction’ — is the natural semantics given the attested body-part function of the forms.

4. The Heteroclititic Gift-Noun

The δωρ-/δαρ- alternation and the Celtic u can be addressed together, through Clayton’s (2023) reconstruction of the PIE *-wṛ-/-(w(e/o)n- heteroclite class.

Clayton’s dissertation systematically reconstructs a class of neuter heteroclitites with alternating strong stems in *-wṛ- and weak stems in *-w(e/o)n-, and demonstrates that a regular metathesis *wṛ > *ru operated at Proto-Nuclear-IE depth — after the Anatolian split, before branch separation. Critically, Clayton independently identifies *deh₃- as a root that forms heteroclitites of

this class, with Sanskrit *dārú-* ‘liberal’ (from **déh₃-w_ṛ-*) and Prasun *pyörü* ‘gift’ (from **pro-deh₃-w_ṛ-*) among his examples. His argument is not concerned with Celtic or with the etymology of *δῶπον*; the application to those forms is the contribution of the present paper.

With this framework in place, the derivation of *δῶπον* from the heteroclitic strong stem is as follows: PNIE **deh₃-w_ṛ-* undergoes the regular metathesis to give **deh₃-ru-*; the laryngeal **h₃* colors and lengthens the preceding vowel, giving **dō-ru-*; thematization yields **dōr-o-n-*; Greek surfaces as *δῶπον*. The standard exemplar of the **w_ṛ > *ru* metathesis is **k^wetw_ṛ-* ‘four’ → **k^w(e)tru-* in Avestan *caθru-*, Gaulish *petru-*, Latin *quadru-* (Clayton 2023: §2.2). The mechanism is well-attested and uncontroversial; its application to **deh₃-* follows from Clayton’s independent identification of this root as a member of the relevant class.

The weak stem accounts for the remaining Greek and Latin forms. The heteroclitic weak stem **deh₃-w(o)n-* (zero-grade **dh₃-w(o)n-*) yields Greek *δῶνος* ‘gift, loan’ with regular zero-grade vocalization. Latin *dōnum* is the same weak stem, thematized, with the long *ō* from laryngeal coloring in the full grade. The *δωρ-/δαρ-* alternation that Schrijver described as ‘obscure’ is the strong/weak stem contrast of a single heteroclitic paradigm.

Form	Language	Stem	Analysis
<i>δῶπον</i>	Greek	strong, metathesized	<i>*deh₃-w_ṛ- > *dō-ru-</i>
<i>δῶνος</i>	Greek	weak, zero-grade	<i>*dh₃-w(o)n-</i>
<i>dōnum</i>	Latin	weak, full grade, thematized	
<i>dārú-</i>	Sanskrit	strong, metathesized	Clayton (2023)
<i>*durno-</i>	Celtic	strong, metathesized + revocalized	§5 below
<i>dūre</i>	Latvian	strong, Baltic development	Schrijver (1995)

Table 2. The gift-noun cognate set under the heteroclitic analysis.

5. Celtic **durno-* and the Metathesis Pathway

The Celtic derivation requires separate treatment because of the original *u* and because the compound evidence bears on the base meaning.

As noted above, the *u* vocalism is attested in Continental Gaulish from the first century BCE — *Durnomagus*, *Durnacos*, *Dagodurnus* — predating any Brythonic sound change. The chain from PNIE **deh₃-w_ṛ-* to Celtic **durno-* runs as follows: the PNIE metathesis **w_ṛ > *ru* gives

*deh₃-ru-; in non-final syllables adjacent to a labial *w, the syllabic resonant *r̥ is u-colored, yielding *dur- (the Lubotsky mechanism, documented for Greek and Indo-Iranian in Lubotsky 1994, 1997); a singulative suffix *-no- then gives *durno-, after loss of the laryngeal in pre-Celtic. The closest phonological analog in the wider literature is OKhotanese durna-, which Bailey (1961) traces from PIIr. *druna- by the same metathesis and revocalization; Clayton (2023) cites this parallel for his Sanskrit data, and it is equally the best available analog for the Celtic development.

The compound evidence bears on the base meaning. Proto-Celtic *are-durnon (Old Irish aurdorn, Welsh arddwrn, Breton arzorn) means ‘wrist’; *ambi-durnos (Old Irish imdorn) means ‘sword hilt.’ Neither compound is semantically natural with ‘fist’ as the base; both are natural with ‘hand’ construed as an anatomical unit. Breton reinforces this reading synchronically: dorn forms the body-part dual daouarn (= daou ‘two’ + dorn), a category Breton reserves for anatomically paired body parts (Ball & Müller 2009: 445–446). The derivation dorn → dornad ‘handful’ (with the suffix marking container contents) likewise presupposes an open-hand reading of the base. The semantic evidence from Celtic compounds and derivatives is consistent with a body-part noun meaning ‘hand’ rather than a posture term meaning ‘fist,’ which aligns with the reconstruction from *de-r- as ‘at the place of the hand-direction.’

6. The δωρ-/δαρ- Alternation and the Vitruvius Testimony

The alternation between δωρ and δάρις — documented by Schrijver as formally obscure — becomes transparent under the heteroclitlic analysis: δωρ is the thematized strong stem (*deh₃-wr̥- > *dōru-) and δάρις is an i-stem derivative of the bare body-part noun (*de-r-). They are not the same surface formation and need not exhibit the same vocalism; the connection is family relationship through the shared pre-PIE base.

Beekes’s decision to separate the ‘gift’ and ‘palm-breadth’ senses of δωρ into unrelated entries addresses a real formal problem. The framework offers an alternative: both senses descend from the same hand-root, one through the gift-noun branch and one by family association with the body-part noun. That the two senses were understood as connected in antiquity is confirmed by Vitruvius, who states explicitly that the Greeks call the palm δωρ because giving (datio) is called δωρ in Greek, and a gift is always conveyed through the palm of the hand (De architectura 2.3.3). Vitruvius’s testimony does not settle the etymological question — an ancient author’s folk etymology is not a formal derivation — but it is evidence that the semantic connection between the two senses was transparent to Greek speakers, which is consistent with a shared etymological origin rather than accidental polysemy.

7. Previous Accounts

Pokorny’s assembling of the cluster at IEW 203 was the essential first step; the difficulty was not in recognizing the forms as related but in providing the formal account. The framework supplies what the IEW entry left open: the $\delta\omega\rho$ -/ $\delta\alpha\rho$ - alternation as a heteroclitic strong/weak contrast, and the Celtic *u* as a metathesis reflex.

Schrijver’s (1995: 58) assessment — ‘obscure’ — accurately described the state of the evidence. The framework does not so much correct Schrijver as supply the formal tool that was missing: the Clayton (2023) heteroclitic framework was not available at the time. Schrijver’s observation that Celtic *u* is original is one the present analysis depends on and confirms.

Matasóvič’s (2009) substrate hypothesis followed from the absence of a viable IE etymology, which is the appropriate methodological response. The claim of the present paper is that a viable IE etymology is now available; if that claim is correct, the substrate hypothesis is no longer necessary as an explanatory device. Whether it is correct is a question for the specialist community.

Beekes’s separation of the two senses of $\delta\omega\rho\omicron\nu$ and his exclusion of Celtic on grounds of *u*-vocalism are both formally motivated within the evidence available to him. The framework offers an alternative account under which the two senses are related and the Celtic *u* is not a counterargument but an expected outcome. The two accounts make different predictions about the morphological history and can in principle be evaluated independently of each other; the present paper offers the framework for that evaluation.

Schrijver (1995: 58), citing Beekes, also noted parenthetically that ‘in the Greek forms, δ - might reflect $*du$ - (cf. $\delta\omega\delta\epsilon\kappa\alpha$ ‘12’ < $*du\check{\sigma}$ -)’ — an observation that anticipates the morphological-identity argument developed in Paper 2, where the numeral $*du\omicron$ - is reconstructed as a derivational sibling of the contrastive suffix $*(t)ero$ - on the same root.

8. Anatolian Evidence and Chronological Architecture

Independent semantic evidence from Anatolian supports the chronological architecture implied by the heteroclitic analysis. Norbruis (2021: 243–249) argues, on Anatolian-internal grounds, that the original meaning of $*deh_3$ - was ‘to take’ rather than ‘to give’: the give-meaning is a post-Anatolian innovation. His reconstructed semantic development — ‘to take’ > (with a goal or beneficiary constituent) ‘to bring, convey, deliver’ > (when the recipient is animate) ‘to hand over, to give’ — is independently motivated by comparative evidence including the Prestino inscription (Eska 2004), which attests the intermediate stage of physical delivery to divine recipients. Norbruis’s argument is independent of the present project and does not require any of its proposals.

This bears on the heteroclititic gift-noun in the following way. If the give-meaning is a post-Anatolian innovation, and if the gift-noun family (*deh₃-wr-) is itself post-Anatolian — as Clayton’s PNIE metathesis, which by definition postdates the Anatolian split, requires — then the chronological layering is coherent: body-part noun first (pre-Anatolian), directional particle second (pre-Anatolian), take-verb third (pre-Anatolian, Anatolian branch), give-verb and gift-noun fourth (post-Anatolian, PNIE). The absence of the gift-noun family from Anatolian is expected, not accidental. Paper 4 develops the formal account of Hittite dā- ‘to take’ as the Anatolian representative of the take-verb branch.

9. Scope and Limitations

Several clarifications are necessary to define the proposal’s scope precisely.

The framework does not introduce a new PIE root. The bare *de- and the locative *-r- are independently attested morphemes; the proposal concerns their prior productive relationship. A reviewer committed to atomistic root reconstruction will resist the compositional analysis on principled grounds, and that resistance is coherent; the framework’s claim is that the compositional reading is explanatorily superior to treating each reflex independently.

The reconstruction of *deh₃- as the correct post-Anatolian verbal pre-form is not in dispute. The claim is about the internal morphological structure of that form — that *-h₃- is a derivational extension suffix rather than an unanalyzable root consonant — and about the chronology of its addition. The phonological evidence for *h₃ in the post-Anatolian period is accepted; the question is only when and how the laryngeal was added.

The framework does not claim that all IE ‘hand’ words descend from *de-r-. PIE *ǵh₁esr- (Greek χεῖρ, Hittite keššar) is a separate hand-word with its own well-attested comparative set, and the present analysis makes no claim about it.

The older derivation of δῶρον from *deh₃-ro- remains formally available. Clayton’s framework is adopted here because it accounts for the Celtic cognates that the older derivation could not, but specialists may reasonably prefer the older analysis pending further evaluation of Clayton’s reconstruction.

Latvian dūre is registered as supporting evidence but is not load-bearing; the Greek and Celtic evidence is sufficient to ground the lexical claim independently.

10. The Remaining Papers

Paper 2 (‘A Better Hand’) reconstructs the PIE contrastive suffix **(t)ero-* compositionally as **de-* + **-r-* + **-o-*, and develops the numeral **duo-/dui-* ‘two’ as a parallel derivational outcome on the same root. The bilateral character of **(t)ero-* — its consistent function of marking ‘one of a pair’ across the family — is argued to be a morphological inheritance from the **-u-* element of pairedness, not a typological consequence of bilateral body symmetry. Dunkel’s (2014) published etymology unifying the prefixal **ui-* and the numeral **dui-*, together with his reconstruction of the inherited compound **ui-tero-* (Vedic *vítarām*, OCS *vŭtoru* ‘the second’), provides the external support.

Paper 3 (‘The Bare Hand’) establishes **de-* as an independently attested PIE directional particle and decomposes the give-verb **deh₃-* as **de-* + verbal extension **-h₃-*. The strongest evidence for this bimorphemic decomposition is a class-by-class asymmetry in the Greek root-aorist 2pl imperative paradigm: roots the framework analyzes as bimorphemic (including **deh₃-* → *δότε*) preserve a bare-CV form in the 2pl, while roots the framework does not analyze as bimorphemic (including **steh₂-* → *στῆτε*) preserve the laryngeal-conditioned long vowel throughout. The asymmetry is descriptively established in the standard grammatical tradition; the framework provides a structural account of why it should exist.

Paper 4 (‘The Hand That Takes’) derives Hittite *dā-* ‘to take’ from a pre-Anatolian directive univerbation **de + a*. The proposal fits a documented productive class in attested Hittite: the Set 1 directional system in which a local element is followed by the directional particle **-a* and contracts to a fixed adverb or preverb (*anda* < **en + a*, *katta* < **kmt + a*, *parā* < **pro + a*). The proposal resolves the take/give polarity — the fact that Anatolian’s reflex of **deh₃-* means ‘take’ while post-Anatolian reflexes mean ‘give’ — as a structural consequence of the directional **a* ‘toward the speaker’ combining with the hand-element to yield ‘extend toward oneself’ = ‘take.’

11. Probability Assessment

We assess the probability of the Paper 1 thesis — that the comparative cluster reflects two derivational siblings on the same pre-PIE hand-element **de-* — at approximately 30–40%, conditional on its own evidence considered independently of Papers 2–4.

In favour of the analysis: the Clayton (2023) framework provides an independently motivated formal mechanism that was previously unavailable, and the present paper’s contribution is its application to the Celtic and Greek data. The OKhotanese *durna-* parallel (Bailey 1961, cited in Clayton) is the closest phonological analog for the Celtic development in the literature. The unification of *δωρ-* and *δαρ-* as strong and weak stems of a single heteroclitic paradigm provides a structural account where Schrijver’s label ‘obscure’ has stood for thirty years. Against: the

compositional body-part noun *de-r- is non-standard; the Celtic derivation involves more steps than earlier proposals; and the broader framework across all four papers requires a productive pre-PIE morphological system whose existence can only be evaluated by assessing the evidence together. The probability at the level of the whole project — integrating the conditional probabilities across Papers 1–4 — is in the range 1–4%, with much of the evidence supporting multiple strata simultaneously rather than each stratum independently. We regard the central claim as more likely wrong than right at the marginal level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

Authorship and Contributions

Chavis’s contributions. The central hypothesis originates with Chavis: the identification of the dual attestation of Greek δῶρον — gift and palm-breadth — as evidence of a single etymological family was the project’s founding insight. Chavis directed the research agenda at every stage: the hypothesis that *-(t)ero- descends from the same root; the identification of ambidexter as the morphologically transparent compound witness for Paper 2; the question ‘whence comes the r in *-tero-?’ that opened the compositional analysis of Paper 3; the hypothesis that Hittite dā- preserves a pre-Anatolian construction; and the decision to incorporate Clayton (2023) as the formal linchpin of the lexical argument. All fundamental decisions about scope, framing, and direction across all four papers rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude’s contributions. Claude’s contributions are primarily in source synthesis, exposition, and analytical development. Across the project, Claude located and read the relevant published literature — including Clayton (2023) on the heteroclite class; Lipp (2009), Bauhaus (2019), and Kahl (2024) on the locative *-r-; Norbruis (2021) on the semantic chronology of *deh₃-; Stifter (2022) on the Bronze of Novallas; Lubotsky (1994, 1997) on the phonological mechanisms; Dunkel (2014) on the *ṽi-tero- compound; Kloekhorst (2008, 2013) and Hoffner & Melchert (2008) on Hittite; Lejeune (1961) and van Hes (2021) on Mycenaean -δε; and ancillary sources throughout. Claude drafted the prose of all four papers from Chavis’s structural and substantive direction; identified Bauhaus’s locative-directive paradigmatic pairs as published structural precedent for Papers 3–4; and integrated the architectural alignment across all four papers at each revision stage. The probability estimates in all four papers are Claude’s own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The

argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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